

Q.1	Q.2	Q.3	Q.4	Q.5	TOTAL
-----	-----	-----	-----	-----	-------

Q.1) (10) Write which of the following is an example of structured data, unstructured data, or semi-structured data.

A JSON file	<i>semi-structured</i>
A CSV file	<i>structured</i>
A tweet	<i>unstructured</i>
An email	<i>semi-structured</i>

Q.2) (30) You are working as a data scientist for a healthcare company. The company has collected a dataset containing various features of patients, such as age, blood pressure, cholesterol levels, and other medical history details. The goal is to predict whether a patient is at risk of developing heart disease (a binary outcome: "Yes" or "No").

Patient ID	Age	Gender	Blood Pressure	Cholesterol	Smoking	Physical Activity	BMI	Diabetes	Family History	Heart Disease
1	45	M	130	200	No	Yes	25.3	No	Yes	No
2	50	F	120	210	Yes	No	27.6	Yes	Yes	Yes
3	39	M	140	180	No	Yes	22.9	No	No	No
4	55	F	150	230	Yes	No	30.1	Yes	Yes	Yes
5	60	M	135	195	Yes	Yes	28.4	No	Yes	Yes
6	52	F	110	205	No	Yes	24.2	Yes	No	No
7	47	M	145	220	Yes	No	26.7	No	Yes	Yes
8	48	F	125	190	No	Yes	23.5	No	No	No
9	63	M	160	240	Yes	No	29.8	Yes	Yes	Yes
10	40	F	118	185	No	Yes	21.9	No	No	No

Column Descriptions:

PatientID: Unique identifier for each patient.

Age: Age of the patient.

Gender: Gender of the patient (M for Male, F for Female).

BloodPressure: Blood pressure measurement of the patient.

Cholesterol: Cholesterol level of the patient.

Smoking: Indicates if the patient is a smoker (Yes or No).

PhysicalActivity: Indicates if the patient engages in regular physical activity (Yes or No).

BMI: Body Mass Index of the patient.

Diabetes: Indicates if the patient has diabetes (Yes or No).

FamilyHistory: Indicates if there is a family history of heart disease (Yes or No).

HeartDisease: Target variable indicating if the patient is at risk of developing heart disease (Yes or No).

The sample data is provided to help you better understand the problem.

Please, do NOT develop a solution that is specific to the sample data provided above!

Based on the given case study, which **supervised learning algorithm** would be most suitable for predicting whether a patient is at risk of developing heart disease? Explain why the chosen algorithm is appropriate for this prediction task. Include the key characteristics of the algorithm and how it fits the nature of the problem.

The most suitable supervised learning algorithm is the Logistic Regression algorithm for the given problem. Logistic Regression is designed for binary classification problems, where the outcome variable is categorical and has two possible outcomes. Logistic Regression provides probabilities for each class label, which is useful in medical predictions as it allows for interpreting the confidence of the model in its predictions. The model coefficients in Logistic Regression indicate the importance and impact of each feature on the prediction. This is valuable in healthcare, as it can help identify which factors (e.g., age, blood pressure, cholesterol) contribute most to the risk of developing heart disease.

While Logistic Regression is suitable, other algorithms like Decision Trees, Random Forests, or Gradient Boosting Machines (GBMs) could also be considered.

MAKEUP EXAM

(This is a 100-minute exam)

Q.3) (20) You are a data scientist for a network security company. The company wants to detect unusual patterns in network traffic that could indicate security breaches or attacks. The dataset includes features such as source IP, destination IP, protocol, bytes transferred, and connection duration:

ConnectionID	SourceIP	DestinationIP	Protocol	Bytes Transferred	Connection Duration
1	192.168.1.1	192.168.1.100	TCP	1500	30
2	192.168.1.2	192.168.1.101	UDP	500	10
3	192.168.1.3	192.168.1.102	TCP	10000	300
4	192.168.1.4	192.168.1.103	UDP	200	5
5	192.168.1.1	192.168.1.100	TCP	2500	60
6	192.168.1.2	192.168.1.101	UDP	150	8
7	192.168.1.3	192.168.1.102	TCP	12000	400
8	192.168.1.4	192.168.1.103	UDP	180	4
9	192.168.1.1	192.168.1.100	TCP	300	15
10	192.168.1.2	192.168.1.101	UDP	1000	25
11	192.168.1.3	192.168.1.102	TCP	15000	500
12	192.168.1.4	192.168.1.103	UDP	50	2

Column Descriptions:

ConnectionID: Unique identifier for each connection.

SourceIP: IP address of the source.

DestinationIP: IP address of the destination.

Protocol: Protocol used for the connection (e.g., TCP, UDP).

BytesTransferred: Number of bytes transferred during the connection.

ConnectionDuration: Duration of the connection in seconds.

The sample data is provided to help you better understand the problem.

Please, do NOT develop a solution that is specific to the sample data provided above!

Given the following sample network traffic data, develop a strategy to detect anomalies. Choose an appropriate anomaly detection algorithm, justify your choice, and describe the steps you would take to implement it. Discuss how you would evaluate the performance of your anomaly detection model.

Isolation Forest is specifically designed for anomaly detection. It works by isolating observations by randomly selecting a feature and then randomly selecting a split value between the maximum and minimum values of the selected feature. It is computationally efficient and can handle large datasets with high-dimensional features, making it suitable for real-time network traffic monitoring. It does not make any assumptions about the distribution of the data, which is advantageous given the diverse and potentially unpredictable nature of network traffic. The algorithm provides an anomaly score for each observation, which helps in understanding how anomalous each connection is.

We use the trained model to assign an anomaly score to each connection in the test set. We need to determine a threshold for the anomaly score to classify a connection as anomalous or normal. This can be based on the contamination rate or chosen to balance precision and recall. We use metrics such as precision, recall, F1-score, and ROC-AUC to evaluate the performance. These metrics help in understanding the trade-off between detecting true anomalies and avoiding false positives. We use cross-validation to ensure the model's robustness and generalizability.

Q.4) (20) You are a data scientist for an online retail company specializing in electronics and gadgets. The company wants to leverage customer purchase data to improve marketing strategies and personalize the shopping experience. The dataset includes features such as customer ID, product categories purchased, purchase frequency, and total spending:

CustomerID	Category1	Category2	Category3	Category4	Frequency	TotalSpending
1	Electronics	Books		Sports	5	1500.00
2	Electronics		Kitchen		3	800.00
3	Books	Electronics		Toys	7	2500.00
4	Electronics	Toys		Sports	4	1200.00
5	Kitchen	Electronics		Toys	6	1800.00
6	Electronics		Sports		2	500.00
7	Books	Toys		Electronics	8	3000.00
8	Electronics		Books	Toys	5	2000.00
9	Electronics	Sports		Books	3	1000.00
10	Sports	Books		Electronics	4	1500.00

The sample data is provided to help you better understand the problem.

Please, do NOT develop a solution that is specific to the sample data provided above!

Develop a recommendation system to suggest new products to customers based on their segment. Describe how you would implement this recommendation system.

We can use a combination of collaborative filtering and customer segmentation techniques to develop a recommendation system for an online retail company specializing in electronics and gadgets.

We ensure there are no missing or inconsistent values in the dataset. We create binary features for each product category to represent whether a customer has purchased from that category. Then we normalize the Frequency and TotalSpending features to ensure they are on a comparable scale.

We use K-Means clustering to segment customers based on their purchase behavior. We use metrics like the silhouette score to determine the optimal number of clusters.

We use collaborative filtering to recommend products within each segment. There are two main types of collaborative filtering:

- i. Recommend products by finding similar customers within the same segment.
- ii. Recommend products by finding items that are like those the customer has already purchased.

MAKEUP EXAM

(This is a 100-minute exam)

Q.5) (20) A retail store has provided you with a sample of their daily sales data for the past week in a CSV file named "sales_data.csv". The data includes columns such as date, product category (category), product ID (product_id), quantity sold (quantity), and total sales amount (sales_amount).

Sample Data (sales_data.csv):

```
date,category,product_id,quantity,sales_amount
2024-05-20,Electronics, TV123, 2, 1200.00
2024-05-20,Clothing, Shirt456, 1, 35.99
2024-05-21,Appliances, Fridge789, 1, 899.99
2024-05-21,Grocery, Milk123, 3, 5.99
2024-05-22,Electronics, Monitor007, 1, 250.00
2024-05-22,Clothing, Pants987, 2, 79.98
2024-05-23,Appliances, Washer345, 2, 599.98
2024-05-23,Grocery, Eggs567, 1, 2.99
```

a) (10) Calculate the average sales amount per day.

```
import pandas as pd

sales_df = pd.read_csv("sales.csv")

daily_sales = df.groupby('date')['sales_amount'].sum()

average_sales_per_day = daily_sales.mean()

print(f"The average sales amount per day is: ${average_sales_per_day:.2f}")
```

b) (10) Filter and display all sales records for the 'Electronics' category.

```
import pandas as pd

sales_df = pd.read_csv("sales.csv")

electronics_data = df[df['category'] == 'Electronics']

print(electronics_data)
```